

Instructor: Date:

Learner: Regional Branch:

Program: Center:

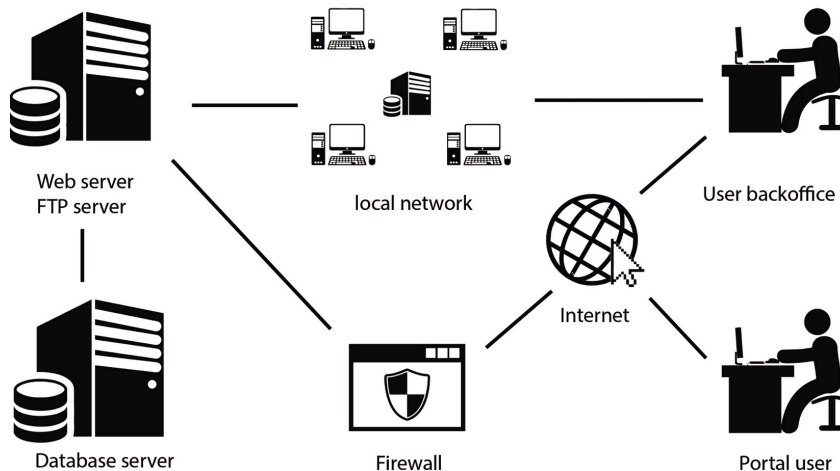
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Let's Start!

The Appearance of the Database Management Systems



Learning Outcome:

Identify information about the history of software and its characteristics in simple spoken and written situations.

Ask your partner:

- How do you think people managed data before computers were invented?
- Do you think a standardized way of managing data makes the use of technology easier? Why? Why not?

Can Do Statements:

- I can talk about the appearance of database management systems.
- I can list some examples of the first database systems.
- I can discuss facts about the appearance of relational database management systems.



Let's Explore!

How did the 1970s innovations influenced* the future generations?

*To "influence" means to have a considerable effect on the way something else happens.



Instructions: Read the text below.

- Frank:** Hello, Mary. I want to know more about database management systems and I understand you are an expert on the subject. Were there any data management models during the first half of the 20th century?
- Mary:** Data management was a big problem during the 1950s. There was no standardized way of storing and accessing data, which means that each program had its own way of managing information.
- Frank:** Were there any innovations that made the data management process easier?
- Mary:** Only until the mid-1960s. It's during this decade that programmers create a basic data access system for space missions. After this, other developers decided to create more sophisticated data models.
- Frank:** How was data managed in these models?
- Mary:** By the end of the 1960s, most models used a hierarchical system that represented data relationships as a tree. This means that for retrieving information, the program has to read the whole tree from the root node.
- Frank:** That doesn't sound easy. I know that one of the most influential innovations took place in 1970, after a researcher wrote a paper that talked about the basis for data management. How were the coming generations influenced by this paper?
- Mary:** This paper proposed a relational database management system. According to this proposal, we store data in tables of related rows and columns.
- Frank:** Was this model called RDBMS?
- Mary:** Exactly. Thanks to this model, we now have standardized data structures. Plus, applications are able to take their information from different sources.
- Frank:** But from what I know, this was a theoretical proposal. Was there someone who put these ideas into practice?
- Mary:** Short after the paper was published, there were successful attempts for making this model for data management come true.



Let's Practice!

Frank, do you have more questions about the beginnings of database management systems?

Instructions: Read the text and circle all the conjugations of verb to be in past.

1. **Frank:** What was the name of the data management system used by John Cullinane in 1968?
Mary: The name was IDMS. It stands for Integrated Data Management Systems.
2. **Frank:** Which was the first attempt to have a standardized way to store data?
Mary: The Generalized Update Access Method (GUAM) was one of the first attempts.
3. **Frank:** Why were the ideas of Edgar Frank Codd important for the history of database management systems?
Mary: Because they were the first step to the Relational Database Management System (RDBMS).



Let's Practice!

The Beginning of Database Management Systems (DBMS)

Instructions: Complete the sentences by writing its corresponding word. Use capital letter where necessary. Use: Was there, Were there, Was there a way, Was there a method, Were there methods. Be careful with the distractors.

1. _____ a standardized way of storing and accessing data before the 1960s? No, there was not.
2. _____ immediate advantages of using a DBMS? Yes, there were. For example, they facilitated the creation of user-oriented languages.
3. _____ of organizing information in a rows and columns before RDBMS? No, RDBMS was the pioneer.



Extension Activity

Let's find more information about Database Managements Systems (DBMS).

Instructions: Complete the following steps. Fill in the Self-Assessment.

- What other first database systems were there in the beginning of database systems? Look for information about it.
- Make a list of the ones you find.
- Write the description of one that you consider important.
- Share your description with a classmate.

Self-Assessment

Yes No

- I look for information about database systems.
- I list some database systems.
- I describe a database system.

☐	☐
☐	☐
☐	☐



Target Words

- **attempt** (noun): an effort for accomplishing or getting something done
e.g. There were successful attempts for making this model for data management come true.
- **relationship** (noun): a connection or relation between two or more elements
e.g. Most models used a hierarchical system that represented data relationships as a tree.
- **root node** (noun): a topmost or bottom node in a tree data structure
e.g. The program has to read the whole tree from the root node.
- **innovation** (noun): something that is new and different from its predecessors
e.g. Were there any innovations that made the data management process easier?
- **manage** (verb): to control or work with something
e.g. How was data managed in these models?
- **sophisticated** (adjective): something that is complex and that has a high quality
e.g. After this, other developers decided to create more sophisticated data models.



Learning Points

Simple Past Tense of the Verb 'to Be' in Questions

We use the **simple past tense** of the **verb to be** to talk about situations in the past. Let's see how to create questions using those two grammatical elements.

When we want to use the **simple past tense** of the **verb to be** for asking questions, we use the following structure:

Verb to be in simple past + Subject + Complement

Example: **Were** the database management systems easy to work with during the 60's?

As you can see in the example above, when we ask a question, we place the verb before the subject.

When we have an information question (or w-h question), we place the interrogative adverb (or w-h word) before the **verb to be**.

Example:

When was the first database management system invented?

Let's take a look at the conjugation of the verb to be:

Subject	Example
I	<i>Was I in the office last night?</i>
You, They, We	<i>Were you organizing data?</i>
He, She, It	<i>Was it difficult to understand the functioning of database management?</i>

Examples

- *How was data managed in these models?*
- *How were the coming generations influenced by this paper?*
- *Was this model called RDBMS?*





Learning Points

There + Verb 'to Be' in Simple Past Tense in Questions

We use **there was** and **there were** to talk about the past existence of something.

When we want to know if something existed or not in the past, we can use **there was** or **there were** in the interrogative form.

Example:

Were there many database models after the 1970s?

Remember that when we ask a question, we place the verb before the subject.

It is also important to remember that we use **there was** when the subject is singular.

For example:

Was there a standardized model for database management?

We use **there were** when the subject is plural.

For example:

Were there several standardized models for database management?

Examples

- *Was there someone who put these ideas into practice?*
- *Were there any data management models during the first half of the 20th century?*
- *Were there any innovations that made the data management process easier?*